BRAC UNIVERSITY

Department of Computer Science and Engineering

CSE 341: Microprocessors Semester: Summer 2026

Assignment 04 Total Marks: 40

Question 1 [15 marks]

	Instruction
i)	`IN AX, 2A40H`
ii)	`MOV DX, 2A40H` `IN AX, DX`
iii)	`MOV AX, [2A40H]`

Here, **2A40H** is an I/O port number.

a) Identify **addressing technique** of each instructions mentioned above and justify your answer based on the **8086 instruction format** and **addressing constraints. [2]**

b) Deduce if any of these instructions are **invalid or not.** Provide your reasons. **[1]**

c) *"The I/O instruction technique showcased in the 3rd instruction can access a greater number of ports compared to the one in the 2nd instruction."* - Critically evaluate the statement with proper technical reasoning. **[2]**

d) A Computer System uses a modified DMA controller named **DMA6,** which supports **6 DMA channels** instead of the conventional 4-channel architecture of the 8237 DMA controller.

   i) Construct a **labelled diagram** showing how **two secondary DMA6** controllers can be interfaced with one primary DMA6 controller in **cascading mode** using **appropriate pins. [2]**

   ii) Explain the **sequence of operations** involved during **DMA-based data transfer** between the Intel 8086 microprocessor and the **cascaded DMA6 controllers. [2]**

   iii) Describe the values of the given pins of an 8237 DMA Controller during a **DMA write cycle:** *HRQ, HLDA, DREQ, DACK, IOR', IOW', MEMW',* and *MEMR'*. **[3]**

e) For the modified DMA controller named **DMA6** mentioned in **question-(d),** answer the following questions:

i) To adopt the utility of **8 channels**, calculate the **minimum** number of secondary **DMA6** controllers needed. **[1]**

ii) Suppose, **DREQ5** and **DACK5** pins are **faulty.** Now, calculate the **minimum** number of secondary **DMA6** controllers needed to adopt the utility of 8 channels. **[1]**

iii) Suppose, **DREQ1** and **DACK5** pins are **faulty.** Will the number of secondary **DMA6** controllers needed to adopt the utility of 8 channels change? **[1]**

Question 2 [13 marks]

Address	Data
0006CH	00H
0006DH	03H
0006EH	00H
0006FH	05H
0009CH	00H
0009DH	07H
0009EH	00H
0009FH	00H
00108H	00H
05300H	89H
05301H	8BH
00700H	2BH
00701H	90H

a) Use the provided table to calculate the **starting address** of the ISR for interrupt **type 27.** Also, deduce the **first 2-byte hex instruction** that will be fetched to the instruction queue from the ISR. **[2.5]**

b) Explain if the interrupt caused by the `INT 27` instruction is **maskable or not.** Provide your reasoning. **[2]**

c) Suppose the above-mentioned ISR is a program consisting of nothing but **25** MOV operations, an **IRET** instruction at the end, and nothing else. The structure of each MOV operation in the ISR is **MOV DH, [SI + XXXX h],** where **XXXX** is an **Offset.** Calculate the **physical address** where **we can find the IRET instruction. [2.5]**

d) Assume that the **IR6** and **IR7** pins of several 8259 PICs **do not work,** and **CAS2** is always giving a low **(0)** signal. Determine the **maximum number of interrupts** that can be serviced through these **faulty** PICs. Show the necessary calculations and explain your reasoning. **[3]**

e) For a particular project, your microprocessor needs to provide service to **16 interrupts** coming from I/O devices. If you use the **same faulty** PICs mentioned above then explain **how many slave PICs will be required?** Also using the proper **diagram** illustrate how the **master and slave PICs** will be **connected** with each other. **[3]**

Question 3 [3 marks x 4 = 12 marks]

Determine the **number of bus cycles** required for each of the following 8086 I/O instructions and **illustrate timing diagram.** Your timing diagram must include *CLK, AD0-AD15, A16-A19, ALE, RD', WR', M/IO', READY* and *DEN'* pins.

i) OUT AL, 04H

ii) IN AL, 05H

iii) OUT AX, 02H

iv) IN AX, 03H